(3.7) Evaluate $\int_C y\,dx + (x+y^2)\,dy$, where C is the ellipse $4x^2 + 9y^2 = 36$ with counterclockwise orientation. *(5 points)*

$$= \int_0^{2\pi} (2\sin t \cdot -3\sin t\,dt) + ((3\cos t + (2\sin t)^2)(2\cos t))dt$$

$$= \int_0^{2\pi} (-6\sin^2 t + 6\cos^2 t + 8\sin^2 t\cos t)dt$$

$$= \int_0^{2\pi} (6\cos(2t) + 8\sin^2 t\cos t)dt$$

$$= 3\sin 2t \Big|_{t=0}^{t=2\pi} + \int_0^{2\pi} 8\sin^2 t\cos t\,dt$$

Let $u = \sin t$, at $t=0 \Rightarrow u = \sin(0) = 0$

$du = \cos t\,dt$ $\quad t = 2\pi \Rightarrow u = \sin(2\pi) = 0$

bounds are same and equal 0

$$= 3\sin 2t \Big|_{t=0}^{t=2\pi}$$

$$= 0 - 0 = \boxed{0}$$

$$\frac{x^2}{9} + \frac{y^2}{4} = 1$$

$a = 3,\ b = 2$

$x(t) = 3\cos(t)$

$y(t) = 2\sin(t)$

$t\ [0, 2\pi]$

$dx = -3\sin(t)dt$

$dy = 2\cos(t)dt$